DATTA SAI

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Summary

Master's in Computer Science at **University of Illinois Chicago** with nearly 4 years of experience building production grade data and ML systems using **Python**, **SQL**, **PyTorch**, and **TensorFlow**. Expertise in **ETL/ELT**, data modeling, and data quality validation to deliver analytics-ready datasets, plus end-to-end **NLP** and **time-series** model development. Skilled in deploying scalable services via **Docker**-based APIs, with focus on performance and measurable business impact.

Experience

Data Engineer Mar 2025 – Present

University of Illinois Chicago, USA

- Owned end-to-end **ETL/ELT** for **StarREZ** housing data, building automated ingestion and transformation workflows in **Python (Pandas)** and **SQL** to deliver reliable occupancy and turnaround reporting.
- Wrote optimized **SQL** (CTEs, window functions, conditional aggregations) to compute vacancy rate, turn time, SLA compliance, and churn signals for **1,000+ units**, enabling faster operational decisions.
- Built and maintained **Power BI** dashboards and semantic models using **Power Query** and **DAX** (occupancy%, SLA breaches, trend deltas), reducing manual reporting effort by **40%** and improving stakeholder self-serve analytics.
- Implemented **data quality** and reconciliation checks (null/duplicate rules, schema validation, row-count balancing) and maintained version-controlled pipelines in **Git** with documentation for repeatable runs and audit-ready reporting.

System Engineer Aug 2022 – Aug 2024

Alstom Transportation, Bangalore, India

- Built **iSenS**, an **AI Requirements Management** tool for contract engineering, implementing document ingestion and processing pipelines to convert customer specs into structured requirement records for downstream workflows.
- Implemented end-to-end extraction for PDFs and scanned documents using **Python**, **pdfplumber**, **OpenCV**, and **Tesseract OCR**, persisting cleaned text, metadata, and traceable document references in **PostgreSQL**.
- Trained and deployed NLP models to **classify Requirements vs Information** and to **characterize requirements by stakeholder**, using **spaCy**, **Scikit-learn**, and **PyTorch** with feature engineering and evaluation-driven iteration.
- Developed a **RAG chatbot** over Alstom internal railway documents by implementing chunking, embeddings, and **vector search (FAISS)** to answer questions with grounded outputs and precise values pulled from source passages.
- Shipped a production inference layer using **FastAPI** and **Docker** on **AWS** (S3 for storage, CloudWatch for logs), enabling low-latency APIs for requirement classification, stakeholder mapping, and retrieval-based Q&A.

Data Science Intern Jan 2022 – Aug 2022

Ernst & Young, Chennai, India

- Built **ETL/ELT** data pipelines to extract incident/ticket data from **ServiceNow** and internal **SQL** sources; performed cleaning, joins, de-duplication, and **schema validation** in **Python (Pandas)**.
- Executed **feature engineering** (lags, rolling aggregates, workload/queue signals, seasonality features) and generated training labels for **time-to-response** and **time-to-resolution** prediction.
- Trained and tuned models in **Scikit-learn** (Linear/Ridge/Lasso, Decision Tree) using **train/validation splits** and **cross validation** evaluated performance with **MAE/RMSE** and error segmentation by priority/category.
- Automated batch inference and reporting by writing predictions back to **SQL** tables and building **Power BI** dashboards with **Power Query + DAX** to monitor SLA trends, bottleneck drivers, and segment-level drilldowns.

Skills

Programming Languages & Tools: Python, SQL, R, Scala, Bash, Git, Jupyter, VS Code, Linux, GitHub Actions
Machine Learning: Scikit-learn, XGBoost, LightGBM, Random Forest, Regression, Classification, Clustering
DL & NLP: PyTorch, TensorFlow, Keras, BERT, GPT, Transformers, LSTM, CNNs, spaCy, Hugging Face, ONNX
Data Engineering: Spark, Airflow, Kafka, Hadoop, Hive, ETL Pipelines, Data Warehousing, PostgreSQL, MySQL
MLOps & Cloud: MLflow, Docker, Kubernetes, CI/CD, FastAPI, AWS, Azure ML, Model Deployment, Terraform
Analytics: Power BI, Tableau, Excel, Matplotlib, Seaborn, Plotly, A/B Testing, Statistical Analysis, KPI Dashboards
LLM & GenAI: Hugging Face Transformers, PEFT (LoRA/QLoRA), LangChain, RAG, vLLM, FAISS, LangGraph

Education

University of Illinois Chicago, Chicago, USA — Aug 2024 – May 2026
Master of Science in Computer Science
Relevant Coursework: Data Science, AI Engineering, Network Communication, System Engineering, Machine Learning

SRM University, Chennai, India — June 2018 – June 2022
Bachelor's of science in Computer Science
Relevant Coursework: Software Engineering, Data Structures, Operating Systems, Object Oriented Programming

Projects

Real-Time Deepfake Detection | *Python, PyTorch, ResNeXt-50, MTCNN, OpenCV, NumPy, Docker, AWS, Streamlit*

- Built a video-level deepfake detection pipeline by sampling frames, extracting faces with **MTCNN**, and generating embeddings using **ResNeXt-50** for manipulation detection.
- Implemented multi-strategy inference (evenly spaced, early-window, random sampling) and combined outputs via ensemble voting to improve stability across varied video lengths and quality.
- Trained and evaluated classifiers in **PyTorch** with consistent preprocessing, threshold calibration, and confusion matrix driven error analysis to reduce false positives on low-light/low-resolution clips.
- Deployed a GPU-enabled inference app using **Streamlit + Docker** on **AWS**, supporting user uploads, batch inference, and latency-optimized feature extraction.

Lakehouse Predictive Maintenance | *Python, SQL, Spark, Airflow, Parquet/Delta Lake, dbt, MLflow, FastAPI*

- Designed a **lakehouse** pipeline to ingest equipment sensor time-series, store curated data in **Parquet/Delta Lake**, and publish analytics-ready tables for downstream modeling and dashboards.
- Implemented scalable transformations and feature engineering in **Spark** (lags, rolling stats, utilization signals) with orchestration via **Airflow** DAGs for scheduled backfills and daily refresh.
- Added **data quality tests** and schema validation using **Great Expectations** and transformation using **dbt**.
- Training forecasting / failure-risk models (LSTM and tree-based baselines) with experiment tracking in **MLflow** and deploying batch + online scoring through a **FastAPI** service.
- Built production hooks for monitoring (latency, prediction drift proxies, data freshness) and versioned model releases via containerized **Docker** deployments.

Enterprise RAG Assistant + Low-Latency LLM Serving | *Python, Hugging Face, Transformers, FAISS, vLLM*

- Built a retrieval-augmented generation (**RAG**) system over domain documents using chunking + embeddings and **FAISS** vector search to return grounded answers for one line prompts.
- Implemented an inference gateway with **FastAPI** and streaming responses, added request logging, rate-safe batching hooks, and versioned routing for controlled releases and serving low latency generation using **vLLM** for efficient decoding, with response formatting designed to return exact values and source linked passages.
- Added evaluation and regression tests for answer quality (faithfulness, retrieval hit rate, and task accuracy) and guardrails to reduce hallucinations on high risk queries.
- Built a PEFT based adaptation pipeline (**LoRA/QLoRA**) to specialize an open-source model on domain style prompts with reproducible training/evaluation.

Certifications

- **Microsoft Certified:** Azure AI Engineer Associate
- **NVIDIA-Certified Professional:** Accelerated Data Science
- **Databricks Certified:** Machine Learning Professional

Open Source

UW Data Mosaic (uwdata) mosaic-core — 2026

- Contributed to **UW Data Mosaic (uwdata)** by implementing **latest-only query stream scheduling** in the Coordinator/QueryManager, pruning stale requests during bursty UI interactions to improve time-to-latest-result.
- Produced an evidence-backed audit of Mosaic's end-to-end query path (**Coordinator** to **Connector** to **WebSocket** to **Server** to **DuckDB**) and shipped unit tests to validate deterministic suppression of obsolete results.